

CSA0614

## DESIGN AND ANALYSIS OF ALGORITHMS

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### Experiment 6: Merge Sort

**Code:**

```
def merge_sort(arr):
    if len(arr) > 1:
        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]
        merge_sort(left)
        merge_sort(right)
        i = j = k = 0
        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1
        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1
        while j < len(right):
            arr[k] = right[j]
```

```
j += 1
k += 1
arr = [38, 27, 43, 3, 9, 82, 10]
merge_sort(arr)
print("Sorted Array:", arr)
```

**Output:**

```
===== RESTART: C:/Users/hk838/Downl
Sorted Array: [3, 9, 10, 27, 38, 43, 82]
```